MOHAMMED ADIL

SOFTWARE ENGINEERING LEAD | FULL STACK | AI / ML
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PROFESSIONAL SUMMARY

Software Engineering Lead with 7+ years of experience in full-stack and backend development. Experienced in building enterprise applications with React, Angular, Java, Spring Boot and Python, and applying machine learning, NLP, deep learning and Generative AI to practical healthcare and software engineering problems. Combines software delivery experience with hands-on AI/ML projects, including recommendation systems, predictive modeling and Retrieval-Augmented Generation (RAG).

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, TypeScript, SQL

Frontend: React.js, Angular, Node.js, HTML, CSS

Backend & APIs: Spring Boot, Reactive Spring Boot, Flask, FastAPI, REST APIs, asynchronous Python

Generative AI: LLMs, RAG, advanced RAG, LLM integration, prompt engineering, information retrieval, embeddings, semantic search, LLM evaluation, Agentic AI concepts

Vector Search: ChromaDB, FAISS, vector databases

Machine Learning: Linear Regression, Logistic Regression, Decision Tree Classifier and Regressor, Bagging, Random Forest, AdaBoost, Gradient Boosting, XGBoost, KNN

Deep Learning & NLP: PyTorch, FFNN, CNN, RNN, LSTM, GRU, transformers, text preprocessing, text vectorization, sentiment analysis, sequence processing

Cloud & DevOps: Microsoft Azure, Azure APIs, Databricks, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD

Other: Microservices, Agile, model evaluation, model validation, healthcare AI

PROFESSIONAL EXPERIENCE

UHG – Optum Global Services | Software Engineering Lead | March 2026 – Present

Continuing contributions across full-stack engineering and AI/ML initiatives for healthcare technology products.

UHG – Optum Global Services | Senior Software Engineer | August 2022 – February 2026

- Contribute to healthcare technology products, including the Plan Recommendation Tool (PRT) and Online Enrollment (OLE), across frontend and backend development.
- Built and enhanced plan recommendation capabilities using React and Java/Spring Boot REST APIs, supporting the member journey from preferences to plan recommendations.
- Worked on the Plan Recommendation Engine, including business logic and plan matching; contributed to the move toward an ML-based recommendation approach using a KNN model.
- Developed and supported reactive enrollment services for plan enrollment workflows.
- Led the migration of application screens from Angular to React, helping the team move to a modern frontend codebase.
- Built an admin validation tool that reduced a validation task from around 10 minutes to around 30 seconds using React and Spring Boot APIs.
- Supported Azure deployments and containerized applications using Docker and Kubernetes; worked with CI/CD pipelines and application monitoring tools.
- Worked with cross-functional teams and contributed to engineering practices, technical knowledge sharing and production issue analysis.

Tata Consultancy Services | *System Engineer* | *May 2019 – August 2022*

- Developed and supported enterprise applications across frontend and backend technologies.
- Worked with React and Java-based application development and contributed to application deployments on Azure.
- Used Splunk and Dynatrace to review application logs and support performance and production issue analysis.
- Created and supported test cases and worked with team members to improve application quality.

PERSONAL PROJECTS

Meet Adil AI – Personal Resume Chatbot

Built a personal chatbot to answer questions about my professional experience and skills using my resume as its knowledge source. The project demonstrates a RAG-based approach, LLM integration and a public Streamlit deployment.

GitHub: https://github.com/Adil9978/meet-adil-ai
Live: https://adilconnect.streamlit.app/

RoleMatcherAI – Personal Role Matching Project

Built a personal AI project focused on matching role requirements with a candidate profile and helping explore role fit.

GitHub: https://github.com/Adil9978/RoleMatcherAI

Live: https://adilrolematcher.streamlit.app/

AI DOJO – CAPSTONE PROJECTS | OPTUM TECH UNIVERSITY

Capstone 1 – Clinical Intelligence System: Medical Question Answering Using Advanced RAG
Worked on a medical question-answering platform designed to understand clinical questions, retrieve relevant information from trusted medical knowledge sources and generate clear, context-aware answers grounded in retrieved content. The project focused on natural-language clinical queries, information retrieval, embeddings, semantic search and RAG-based response generation.

Skills: Python, NLP, LLM integration, advanced RAG, information retrieval, vector databases, embeddings, semantic search, Azure Cloud APIs, LLM evaluation.

Capstone 2 – Estimating Treatment Costs Using Predictive Modeling
Worked on predictive modeling for healthcare treatment cost estimation using regression, classification and ensemble machine learning approaches.

Algorithms: Linear Regression, Logistic Regression, Decision Tree Classifier, Decision Tree Regressor, Bagging Classifier, Random Forest Classifier, AdaBoost Classifier, Gradient Boosting Classifier and XGBoost Classifier.

Capstone 3 – Mental Health Concern Classification Using NLP and Deep Learning
Worked on a text classification problem to identify content that may indicate mental health concerns, with a focus on noisy and informal social media text and early triage support. Explored text preprocessing, text-to-number conversion, sentiment analysis and deep learning approaches.

Skills: PyTorch, FFNN regression, CNN classification using MNIST, sequence data preprocessing, RNN classification, LSTM and GRU text classification, transformer-based semantic search.

CERTIFICATIONS & EDUCATION

- AI Dojo Program – Optum Tech University (completed)
- Three AI Dojo capstone projects covering Advanced RAG / Clinical Intelligence, Machine Learning / Predictive Modeling, and Advanced AI/ML / NLP and Deep Learning

Post Graduate Certification Program (1.5 years): Software Engineering for Data Science – IIIT Hyderabad
B.E. – Information Technology: Muffakham Jah College of Engineering and Technology | June 2015 – June 2019